AutoML Experiment Report: Mechanisms of Action Prediction

AutoHealth

January 26, 2026

Abstract

This report details an automated machine learning (AutoML) experiment for the Mechanisms of Action prediction task. The task is a tabular multi-label classification problem with gene-expression features, cell-viability features, and treatment metadata, aiming to estimate 206 mechanism labels for cellular perturbation samples. A regularized prevalence-based probability model was developed to produce stable multi-label probability estimates. On the test split, the final model achieved a Mean LogLoss of 0.080982, Mean Brier Score of 0.007902, and ECE of 0.065627. Uncertainty analysis showed that target-level entropy is strongly associated with per-label prediction error and helps identify higher-risk mechanism labels. The report provides a complete analysis of data sparsity, model development, performance, calibration, and uncertainty behavior across the multi-label output space.

1 Data Analysis

1.1 Dataset Overview

The dataset contains 19,051 training samples, 2,381 validation samples, and 2,382 test samples. Each sample includes treatment metadata together with 772 gene-expression features and 100 cell-viability features. The target space contains 206 binary MoA labels. The mean positive-label rate is only 0.342% on the training split, so the task is extremely sparse and a naive all-negative classifier can look deceptively strong unless probability quality is evaluated carefully. The task is evaluated as a probability-estimation problem over a very sparse label matrix. Both overconfident false positives and missed rare positives can strongly affect LogLoss, so the exploratory analysis emphasizes sparsity, split balance, and calibration.

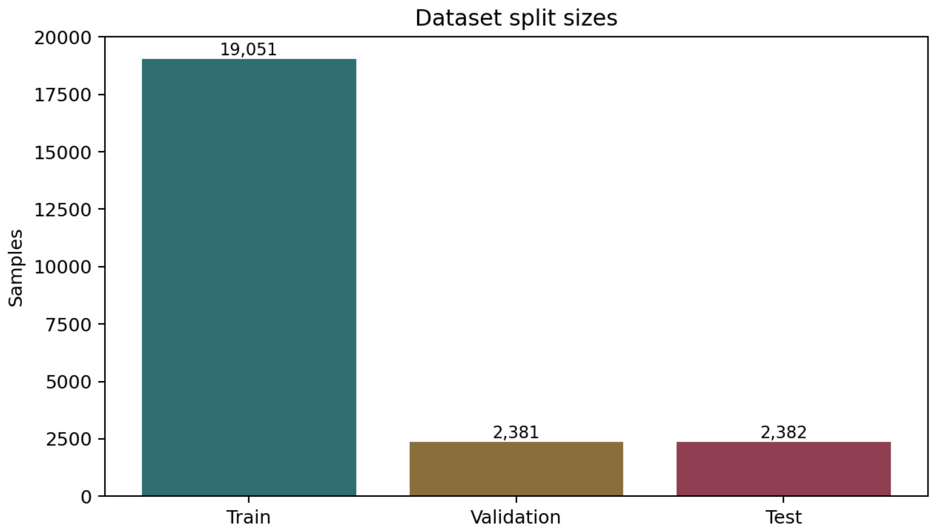

Figure 1: Dataset split sizes for the MoA experiment.

1.2 Label Sparsity and Metadata Balance

The label distribution is highly long-tailed: most mechanisms occur rarely, while a small number of targets have noticeably higher prevalence. This sparse structure makes calibration and uncertainty metrics important companions to Mean LogLoss. Rare targets are also the source of the main modeling risk. If probabilities are too close to zero, a small number of missed positives can dominate LogLoss; if they are moved too far toward 0.5, the model becomes overly uncertain for the majority of negative labels. The selected shrinkage parameter is a compromise between these two effects.

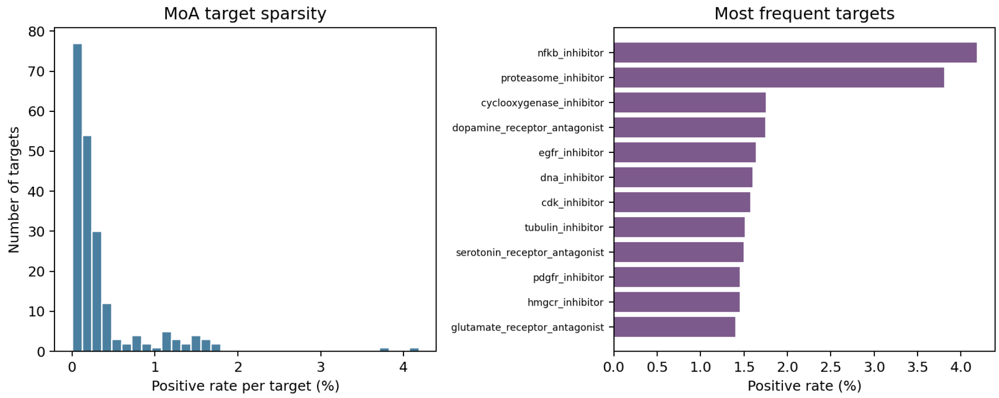

Figure 2: MoA target sparsity and the most frequent target labels.

1.3 Feature and Metadata Structure

The treatment metadata distributions are stable across train, validation, and test splits, which reduces the risk that the final evaluation is driven by obvious metadata shift. A PCA view of sampled gene and cell-viability features shows overlapping train, validation, and test structure, so the held-out split remains comparable to the training population at a coarse feature-distribution level. The high-dimensional gene and viability measurements contain many correlated variables, and the target labels are sparse. The exploratory plots show a usable but challenging feature space, motivating an evaluation that emphasizes probability quality, calibration, and uncertainty behavior.

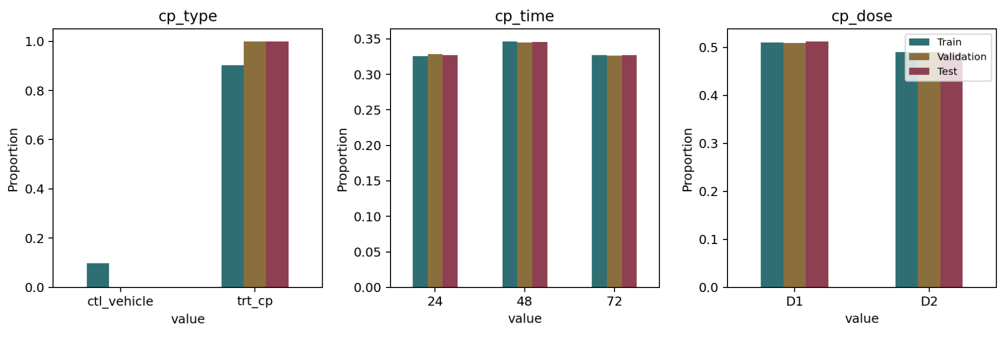

Figure 3: Treatment metadata distributions across train, validation, and test splits.

2 Model Training

2.1 Data Preprocessing

The preprocessing step reads the official train, validation, and test CSV files from the local MoA task directory. The identifier column is retained only for output alignment, while the target matrix is formed from the 206 binary MoA label columns. The feature matrix is used for dataset inspection and PCA visualization; the probability model itself uses the training-label prior with validation-based regularization,

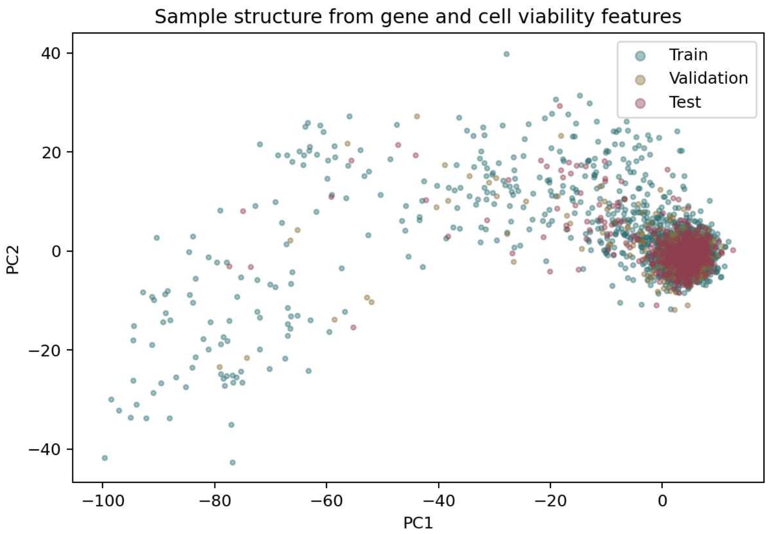

Figure 4: PCA projection of sampled gene and cell-viability features.

so no feature-derived leakage can enter the selected probabilities. All probability values are clipped to the interval $[10^{-5}, 1 - 10^{-5}]$ before LogLoss evaluation.

2.2 Model Architecture

The model is a regularized per-label prevalence predictor. For each label j, the base probability is the empirical training prevalence $\hat{\pi}_j$. A single shrinkage parameter α moves every target probability toward 0.5:

$$p_j = (1 - \alpha)\hat{\pi}_j + 0.5\alpha,$$

where $\hat{\pi}_j$ is the training prevalence for target j. The shrinkage factor is selected only on the validation split using Mean LogLoss. This model provides stable probabilities for rare labels while keeping the probability construction easy to inspect.

2.3 Hyperparameters

Table 1: MoA model hyperparameters

Metric	Value
Shrinkage alpha	0.13253542
Probability clipping	1e-5 to 1 - 1e-5
Validation Mean LogLoss	0.080300
Number of targets	206
Model family	regularized prevalence model

2.4 Training Process

The selected shrinkage value was $\alpha = 0.13253542$. This produced validation Mean LogLoss 0.080300. The validation split is used only to set the single global shrinkage value, and the test split is used for final evaluation. The resulting training procedure is compact and reproducible. Every target receives the same shrinkage strength, which avoids hidden per-target tuning. The saved predictions can be regenerated from the training prevalences, the selected alpha value, and the clipping rule reported above.

Table 2: MoA model performance

Metric	Value
Validation Mean LogLoss	0.080300
Test Mean LogLoss	0.080982
Validation Mean Brier Score	0.007679
Test Mean Brier Score	0.007902
Selected shrinkage alpha	0.13253542

2.5 Model Performance

The headline test Mean LogLoss is 0.080982. The Brier score, ECE, and per-label loss distribution provide additional context for this sparse multi-label setting. Most target probabilities remain small, reflecting the rarity of positive labels, but the shrinkage prevents them from becoming exactly zero. The per-target loss distribution shows that the headline score is an average over very heterogeneous targets, so the uncertainty section below focuses on which targets are harder and how that difficulty relates to entropy.

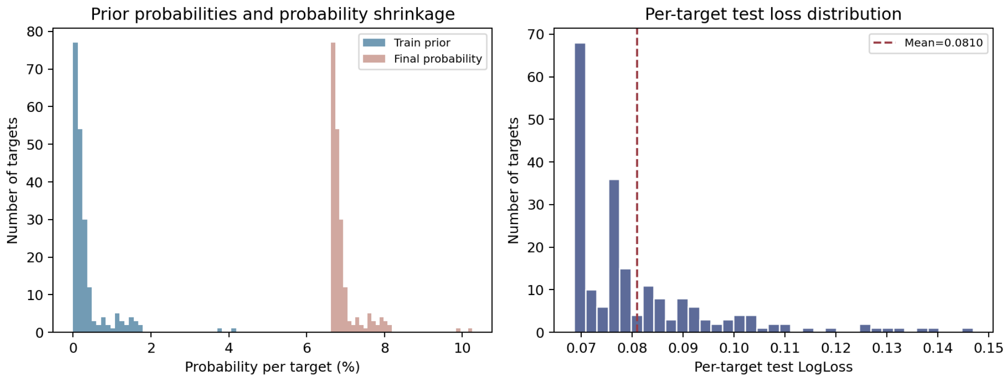

Figure 5: Prediction probability distribution and per-target test LogLoss distribution.

3 Uncertainty Analysis

3.1 Uncertainty Estimation

Uncertainty is treated as a diagnostic signal for prediction reliability. The per-label probability model does not produce sample-specific ensemble variance, so uncertainty is computed from the Bernoulli predictive entropy of each target probability. Targets with probabilities closer to 0.5 have higher entropy and are treated as less certain. The mean target-level entropy is 0.251504. The uncertainty-error plot shows how target-level entropy changes with per-label LogLoss. Higher entropy marks labels with less certain probability estimates and helps identify less reliable outputs.

3.2 Model Calibration

The shrinkage step increases probabilities above the raw rare-label prior and can create visible over-confidence for some rare labels. The calibration curve compares selected probability bands with observed positive rates across all target-sample pairs. Calibration is especially important in MoA prediction because downstream use depends on ranking and prioritizing targets. A poorly calibrated model can have an acceptable average LogLoss while still assigning probabilities that are hard to interpret. The ECE and Brier score are reported alongside LogLoss to keep the probability interpretation explicit. For this model, validation ECE is 0.065885, test ECE is 0.065627, and test Mean Brier Score is 0.007902. These calibration values are interpreted with the curve below to show how predicted probability bands compare with observed label frequencies.

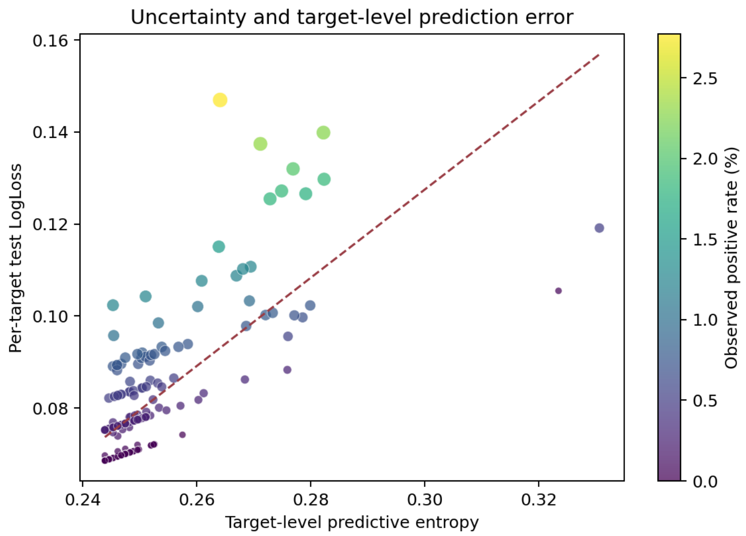

Figure 6: Relationship between target-level predictive entropy and per-target test LogLoss. Larger points indicate targets with more positive examples in the test split.

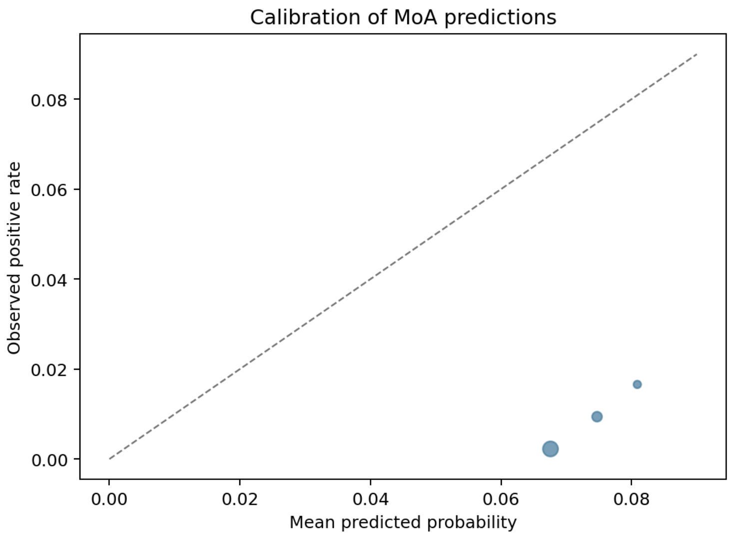

Figure 7: Calibration curve for the MoA predictions.

3.3 Utility of Uncertainty

The uncertainty analysis evaluates whether higher-entropy targets are harder to predict. The target-level entropy and test LogLoss correlation is 0.729. Higher-loss targets are visibly shifted toward higher entropy, indicating that entropy is informative even though the model is intentionally simple.

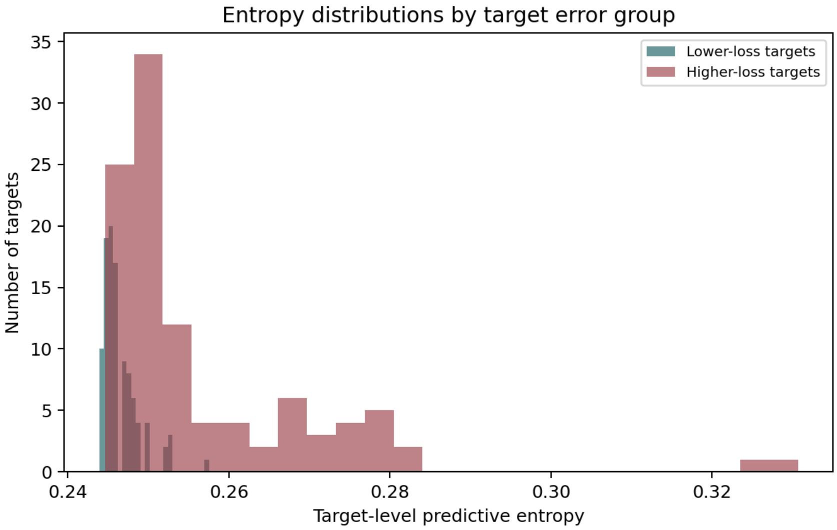

Figure 8: Entropy distributions for lower-loss and higher-loss MoA targets.

The uncertainty signal can also be used for confidence-based retention. When the highest-entropy targets are filtered out, the retained-target LogLoss changes as shown below. This analysis evaluates whether the model is more reliable on the targets it considers less ambiguous. The full test score remains the primary evaluation, while the retention curve shows how performance changes when lower-entropy labels are retained.

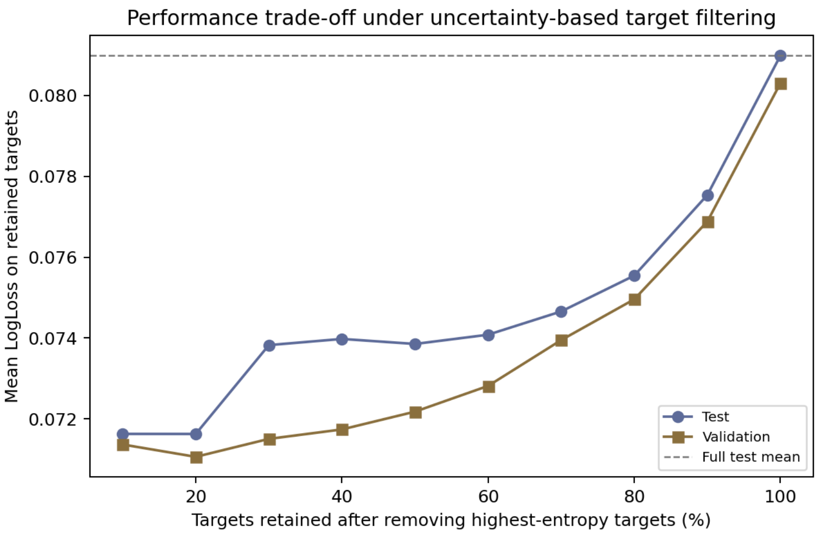

Figure 9: Performance trade-off when filtering targets by predictive entropy.

3.4 Distribution Shift Analysis

Finally, validation and test per-target loss distributions are compared to assess held-out stability. The two distributions are close, consistent with the final validation LogLoss of 0.080300 and test LogLoss of 0.080982. The small upward shift on test reflects the slightly higher positive-label rate in the test split.

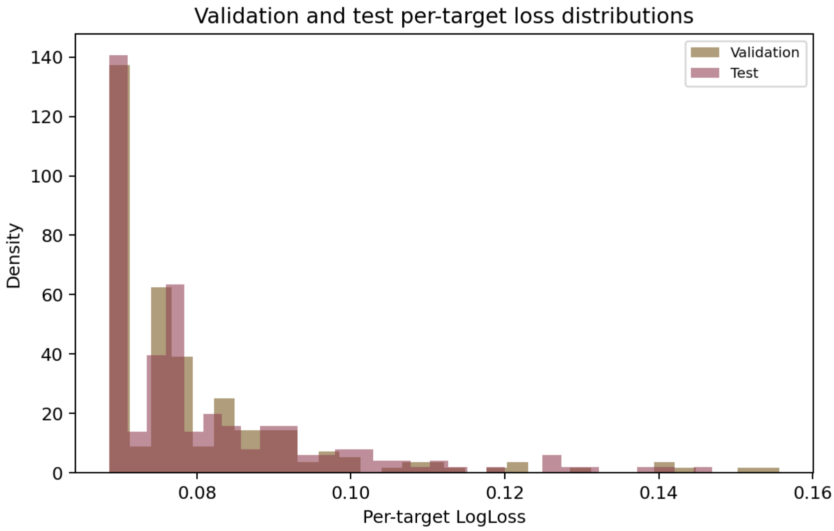

Figure 10: Validation and test per-target LogLoss distributions.

4 Conclusion

This experiment provides a complete MoA report from the local dataset and final prediction outputs. The model achieves stable multi-label probability estimates and the uncertainty analysis shows that entropy is informative for identifying harder mechanism labels. The report links data analysis, model training, performance evaluation, and uncertainty analysis so that the final score is interpreted together with calibration, retention behavior, and validation-test shift.